

## **Procedure: Gaussian Mixture Model (GMM) Using EM Algorithm**

### **1. Initialize the Environment**

- Clear the R workspace and create an output folder.

### **2. Generate Dataset**

- Create synthetic data using Gaussian distributions with different means and covariance matrices.

### **3. Initialize Parameters**

- Set the number of clusters, mixing coefficients, means, and covariance matrices.

### **4. Calculate Gaussian Density**

- Compute the probability of each data point using the Gaussian distribution function.

### **5. Perform E-Step**

- Calculate the responsibility values that represent the probability of each point belonging to a cluster.

### **6. Perform M-Step**

- Update the cluster parameters:
  - Mean values
  - Covariance matrices
  - Mixing coefficients

### **7. Run EM Iterations**

- Repeat E-Step and M-Step until the model reaches convergence.

### **8. Assign Clusters**

- Assign each data point to the cluster with the highest probability.

### **9. Visualize Results**

- Generate cluster plots and log-likelihood convergence graphs.

### **10. Display Results**

- Show final parameters, cluster distribution, and model performance.